

练习题-定积分计算

1. 计算下列定积分

$$(1) \int_0^1 \frac{\ln(1+x)}{(2-x)^2} dx;$$

解

$$\begin{aligned} \int_0^1 \frac{\ln(1+x)}{(2-x)^2} dx &= \int_0^1 \ln(1+x) d\left(\frac{1}{2-x}\right) = \frac{\ln(1+x)}{2-x} \Big|_0^1 - \int_0^1 \frac{1}{2-x} d[\ln(1+x)] \\ &= \ln 2 - \int_0^1 \frac{1}{2-x} \cdot \frac{1}{1+x} dx = \ln 2 - \frac{1}{3} \int_0^1 \left(\frac{1}{1+x} + \frac{1}{2-x} \right) dx \\ &= \ln 2 - \frac{1}{3} [\ln(1+x) - \ln(2-x)] \Big|_0^1 = \ln 2 - \frac{1}{3} (2\ln 2) = \frac{1}{3} \ln 2 \end{aligned}$$

$$(2) \int_0^1 \frac{dx}{(1+x)\sqrt{1+x^2}}.$$

解

$$\begin{aligned} \int_0^1 \frac{dx}{(1+x)\sqrt{1+x^2}} &\stackrel{x=\tan t}{=} \int_0^{\frac{\pi}{4}} \frac{\sec^2 t dt}{(1+\tan t)\sqrt{1+\tan^2 t}} = \int_0^{\frac{\pi}{4}} \frac{\sec^2 t dt}{(1+\tan t)\sec t} \\ &= \int_0^{\frac{\pi}{4}} \frac{dt}{\sin t + \cos t} = \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{4}} \frac{1}{\sin\left(t + \frac{\pi}{4}\right)} d\left(t + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{4}} \csc\left(t + \frac{\pi}{4}\right) d\left(t + \frac{\pi}{4}\right) \\ &= \frac{1}{\sqrt{2}} \ln \left| \csc\left(t + \frac{\pi}{4}\right) - \cot\left(t + \frac{\pi}{4}\right) \right| \Big|_0^{\frac{\pi}{4}} = \frac{1}{\sqrt{2}} \ln(\sqrt{2} + 1) \end{aligned}$$

2. 已知 $f(0)=1, f(2)=3, f'(2)=5$, 计算 $\int_0^1 xf''(2x)dx$.

$$\begin{aligned} \text{解 } \int_0^1 xf''(2x)dx &= \frac{1}{2} \int_0^1 xf''(2x)(2x)' dx = \frac{1}{2} \int_0^1 xf''(2x)d(2x) = \frac{1}{2} \int_0^1 x df'(2x) \\ &= \frac{1}{2} \left[xf'(2x) \Big|_0^1 - \int_0^1 f'(2x) dx \right] = \frac{1}{2} \left[f'(2) - \frac{1}{2} \int_0^1 f'(2x)(2x)' dx \right] \\ &= \frac{1}{2} \left[f'(2) - \frac{1}{2} \int_0^1 f'(2x)d(2x) \right] = \frac{1}{2} \left[f'(2) - \frac{1}{2} f(2x) \Big|_0^1 \right] \\ &= \frac{1}{2} \left[f'(2) - \frac{1}{2} f(2) + \frac{1}{2} f(0) \right] = \frac{1}{2} \left[5 - \frac{1}{2} \cdot 3 + \frac{1}{2} \cdot 1 \right] = 2 \end{aligned}$$

3. 已知 $y' = \arctan(x-1)^2$, $y(0) = 0$, 求 $\int_0^1 y(x)dx$.

解 $\int_0^1 y(x)dx = \int_0^1 y(x)d(x-1)$ (注意, 凑微分 $dx = d(x-1)$ 可以简化计算)

$$= (x-1)y(x)\Big|_0^1 - \int_0^1 (x-1)dy(x) = 0 - \int_0^1 (x-1)y'(x)dx$$

$$= -\int_0^1 (x-1)\arctan(x-1)^2 dx = -\int_0^1 \arctan(x-1)^2 (x-1)d(x-1)$$

$$= -\frac{1}{2}\int_0^1 \arctan(x-1)^2 d(x-1)^2$$

令 $u = (x-1)^2$, 则 $d(x-1)^2 = du$, 当 $x=0$ 时 $u=1$ (积分下限), 当 $x=1$ 时 $u=0$ (积分上限), 于是

$$\int_0^1 y(x)dx = -\frac{1}{2}\int_0^1 \arctan(x-1)^2 d(x-1)^2 = -\frac{1}{2}\int_1^0 \arctan u du$$

$$= \frac{1}{2}\int_0^1 \arctan u du = \frac{1}{2}\left[u \arctan u\Big|_0^1 - \int_0^1 u d \arctan u\right]$$

$$= \frac{1}{2}\left[\frac{\pi}{4} - \int_0^1 u \cdot \frac{1}{1+u^2} du\right] = \frac{1}{2}\left[\frac{\pi}{4} - \frac{1}{2}\int_0^1 \frac{1}{1+u^2} d(1+u^2)\right]$$

$$= \frac{1}{2}\left[\frac{\pi}{4} - \frac{1}{2}\ln(1+u^2)\Big|_0^1\right] = \frac{1}{2}\left[\frac{\pi}{4} - \frac{1}{2}\ln 2\right] = \frac{\pi}{8} - \frac{1}{4}\ln 2$$

4. 计算积分 $\int_0^1 xf(x)dx$, 其中 $f(x) = \int_1^{x^2} \frac{\sin x}{x} dx$.

解

$$\int_0^1 xf(x)dx = \int_0^1 f(x)\left(\frac{x^2}{2}\right)' dx = \int_0^1 f(x)d\left(\frac{x^2}{2}\right) = f(x)\left(\frac{x^2}{2}\right)\Big|_0^1 - \int_0^1 \left(\frac{x^2}{2}\right)df(x)$$

$$= 0 - \int_0^1 \left(\frac{x^2}{2}\right) \cdot f'(x)dx = -\int_0^1 \frac{x^2}{2} \cdot \frac{\sin x^2}{x^2} \cdot 2xdx = -\int_0^1 x \sin x^2 dx = -\frac{1}{2}\int_0^1 \sin x^2 dx^2$$

$$= \frac{1}{2}\cos x^2\Big|_0^1 = \frac{1}{2}\cos 1 - \frac{1}{2}$$

5. 已知 $f(x) = \int_0^x e^{(x-t)^2} dt$, 求 $\int_0^1 [f(x) - f(1)]dx$.

解 令 $u = x-t$, 则 $t = x-u$, $dt = -du$, 当 $t=0$ 时 $u=x$ (积分下限), 当 $t=x$ 时 $u=0$ (积分上限), 所以

$$f(x) = \int_0^x e^{(x-t)^2} dt = \int_x^0 e^{u^2} (-du) = \int_0^x e^{u^2} du$$

于是

$$\begin{aligned}\int_0^1 [f(x) - f(1)] dx &= [f(x) - f(1)]x \Big|_0^1 - \int_0^1 x d[f(x) - f(1)] \\ &= 0 - \int_0^1 x f'(x) dx = - \int_0^1 x e^{x^2} dx = - \frac{1}{2} \int_0^1 e^{x^2} dx^2 = - \frac{1}{2} e^{x^2} \Big|_0^1 = \frac{1}{2} (1 - e)\end{aligned}$$

题 6 需利用公式

$$(1) \int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx$$

$$(2) \int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & f(-x) = f(x) \text{ (偶函数)} \\ 0, & f(-x) = -f(x) \text{ (奇函数)} \end{cases}$$

6. 计算下列定积分

$$(1) \int_{-2}^2 [e^{-x}(1+e^x)^2 \sin^3 x + x^2] dx;$$

解 (关于 $x=0$ 对称区间上的定积分)

$$\int_{-2}^2 [e^{-x}(1+e^x)^2 \sin^3 x + x^2] dx = \int_{-2}^2 e^{-x}(1+e^x)^2 \sin^3 x dx + \int_{-2}^2 x^2 dx$$

对于 $\int_{-2}^2 e^{-x}(1+e^x)^2 \sin^3 x dx$, 令 $\varphi(x) = e^{-x}(1+e^x)^2$, 因为

$$\varphi(-x) = e^x(1+e^{-x})^2 = e^x[e^{-x}(e^x+1)]^2 = e^{-x}(1+e^x)^2 = \varphi(x)$$

所以 $\varphi(x) = e^{-x}(1+e^x)^2$ 是偶函数, 从而 $e^{-x}(1+e^x)^2 \sin^3 x$ 是奇函数, 故

$$\int_{-2}^2 e^{-x}(1+e^x)^2 \sin^3 x dx = 0$$

于是

$$\int_{-2}^2 [e^{-x}(1+e^x)^2 \sin^3 x + x^2] dx = 0 + \int_{-2}^2 x^2 dx = 2 \int_0^2 x^2 dx = 2 \cdot \frac{x^3}{3} \Big|_0^2 = \frac{16}{3}$$

$$(2) \int_{-2}^2 [\sin^3 x + x \ln(1+e^x)] dx;$$

解 (关于 $x=0$ 对称区间上的定积分)

$$\int_{-2}^2 [\sin^3 x + x \ln(1+e^x)] dx = \int_{-2}^2 \sin^3 x dx + \int_{-2}^2 x \ln(1+e^x) dx$$

其中

$$\int_{-2}^2 \sin^3 x dx = 0$$

又利用公式 $\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx$ 得

$$\begin{aligned}\int_{-2}^2 x \ln(1 + e^x) dx &= \int_0^2 [x \ln(1 + e^x) - x \ln(1 + e^{-x})] dx \\ &= \int_0^2 x \ln \frac{1 + e^x}{1 + e^{-x}} dx = \int_0^2 x \ln e^x dx = \int_0^2 x^2 dx = \frac{8}{3}\end{aligned}$$

因此

$$\int_{-2}^2 [\sin^3 x + x \ln(1 + e^x)] dx = 0 + \frac{8}{3} = \frac{8}{3}$$

$$(3) \quad \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[\sin^4 x + x^2 \ln \frac{2-x}{2+x} \right] dx;$$

解 (关于 $x = 0$ 对称区间上的定积分)

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[\sin^4 x + x^2 \ln \frac{2-x}{2+x} \right] dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^4 x dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^2 \ln \frac{2-x}{2+x} dx$$

对于 $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^2 \ln \frac{2-x}{2+x} dx$, 令 $\varphi(x) = \ln \frac{2-x}{2+x}$, 因为

$$\varphi(-x) = \ln \frac{2+x}{2-x} = -\ln \frac{2-x}{2+x} = -\varphi(x)$$

所以 $\varphi(x) = \ln \frac{2-x}{2+x}$ 是奇函数, 从而 $x^2 \ln \frac{2-x}{2+x}$ 是奇函数, 故

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^2 \ln \frac{2-x}{2+x} dx = 0$$

于是

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[\sin^4 x + x^2 \ln \frac{2-x}{2+x} \right] dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^4 x dx + 0 = 2 \int_0^{\frac{\pi}{2}} \sin^4 x dx = 2 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{3\pi}{8}$$

$$(4) \quad \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |\sin x| \arctan e^{-x} dx;$$

解 (关于 $x = 0$ 对称区间上的定积分)

利用公式 $\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx$ 得

$$\begin{aligned}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |\sin x| \arctan e^{-x} dx &= \int_0^{\frac{\pi}{2}} [|\sin x| \arctan e^{-x} + |\sin x| \arctan e^x] dx \\ &= \int_0^{\frac{\pi}{2}} [\arctan e^{-x} + \arctan e^x] \sin x dx\end{aligned}$$

令 $g(x) = \arctan e^{-x} + \arctan e^x$, 则 $g(x)$ 在区间 $(-\infty, +\infty)$ 内可导且

$$g'(x) = \frac{-e^{-x}}{1+(e^{-x})^2} + \frac{e^x}{1+(e^x)^2} = -\frac{e^x}{1+e^{2x}} + \frac{e^x}{1+e^{2x}} = 0$$

所以 $g(x) = C$ ，令 $x = 0$ 得 $C = g(0) = \frac{\pi}{2}$ ，于是

$$\arctan e^{-x} + \arctan e^x = \frac{\pi}{2}$$

因此

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |\sin x| \arctan e^{-x} dx = \int_0^{\frac{\pi}{2}} \frac{\pi}{2} |\sin x| dx = \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \sin x dx = \frac{\pi}{2} \cdot 1 = \frac{\pi}{2}$$

$$(5) \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left[\ln|x + \sqrt{1+x^2}| + \frac{e^x}{1+e^x} \right] \sin^2 x dx.$$

解（关于 $x = 0$ 对称区间上的定积分）

$$\begin{aligned} & \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left[\ln|x + \sqrt{1+x^2}| + \frac{e^x}{1+e^x} \right] \sin^2 x dx \\ &= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \ln|x + \sqrt{1+x^2}| \sin^2 x dx + \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{e^x}{1+e^x} \sin^2 x dx \end{aligned}$$

对于 $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \ln|x + \sqrt{1+x^2}| \sin^2 x dx$ ，令 $\varphi(x) = \ln|x + \sqrt{1+x^2}|$ ，因为

$$\begin{aligned} \varphi(-x) &= \ln|-x + \sqrt{1+x^2}| = \ln \left| \frac{(-x + \sqrt{1+x^2})(x + \sqrt{1+x^2})}{x + \sqrt{1+x^2}} \right| \\ &= \ln \left| \frac{1}{x + \sqrt{1+x^2}} \right| = -\ln|x + \sqrt{1+x^2}| = -\varphi(x) \end{aligned}$$

所以 $\varphi(x) = \ln|x + \sqrt{1+x^2}|$ 是奇函数，从而 $\ln|x + \sqrt{1+x^2}| \sin^2 x$ 是奇函数，故

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \ln|x + \sqrt{1+x^2}| \sin^2 x dx = 0$$

又利用公式 $\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx$ 得

$$\begin{aligned} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{e^x}{1+e^x} \sin^2 x dx &= \int_0^{\frac{\pi}{4}} \left[\frac{e^x}{1+e^x} \sin^2 x + \frac{e^{-x}}{1+e^{-x}} \sin^2 x \right] dx \\ &= \int_0^{\frac{\pi}{4}} \left[\frac{e^x}{1+e^x} + \frac{1}{1+e^x} \right] \sin^2 x dx = \int_0^{\frac{\pi}{4}} \sin^2 x dx = \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2x}{2} dx = \frac{1}{2} \left(x - \frac{1}{2} \sin 2x \right) \Big|_0^{\frac{\pi}{4}} = \frac{\pi}{8} - \frac{1}{4} \end{aligned}$$

因此

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left[\ln|x + \sqrt{1+x^2}| + \frac{e^x}{1+e^x} \right] \sin^2 x dx = 0 + \left(\frac{\pi}{8} - \frac{1}{4} \right) = \frac{\pi}{8} - \frac{1}{4}$$

题 7 需利用公式

$$(1) \int_0^{\frac{\pi}{2}} f(\sin x) dx = \int_0^{\frac{\pi}{2}} f(\cos x) dx \quad (\text{可利用区间翻转变换 } t = \frac{\pi}{2} - x \text{ 证})$$

$$(2) \int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx = \pi \int_0^{\frac{\pi}{2}} f(\sin x) dx \quad (\text{可利用区间翻转变换 } t = \pi - x \text{ 证})$$

$$(3) \int_0^a f(x) dx = \int_0^a f(a-x) dx \quad (\text{可利用区间翻转变换 } t = a-x \text{ 证})$$

$$(4) \int_a^b f(x) dx = \int_a^b f(a+b-x) dx \quad (\text{可利用区间翻转变换 } t = a+b-x \text{ 证})$$

7. 计算下列定积分

$$(1) \int_0^{\frac{\pi}{2}} \frac{\ln(1+\sqrt{\sin x}) - \ln(1+\sqrt{\cos x}) + \sin^3 x}{2} dx;$$

解 由公式 $\int_0^{\frac{\pi}{2}} f(\sin x) dx = \int_0^{\frac{\pi}{2}} f(\cos x) dx$ 得

$$\int_0^{\frac{\pi}{2}} \ln(1+\sqrt{\sin x}) dx = \int_0^{\frac{\pi}{2}} \ln(1+\sqrt{\cos x}) dx$$

所以

$$\begin{aligned} & \int_0^{\frac{\pi}{2}} \frac{\ln(1+\sqrt{\sin x}) - \ln(1+\sqrt{\cos x}) + \sin^3 x}{2} dx \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} [\ln(1+\sqrt{\sin x}) - \ln(1+\sqrt{\cos x})] dx + \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin^3 x dx = 0 + \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3} \end{aligned}$$

$$(2) \int_0^2 \frac{4-x^2}{x+\sqrt{4-x^2}} dx;$$

解 令 $x = 2 \sin t$, 则 $dx = 2 \cos t dt$, 当 $x = 0$ 时 $t = 0$ (积分下限), 当 $x = 2$ 时 $t = \frac{\pi}{2}$

(积分下限), 所以

$$\int_0^2 \frac{4-x^2}{x+\sqrt{4-x^2}} dx = \int_0^{\frac{\pi}{2}} \frac{4 \cos^2 t}{2 \sin t + 2 \cos t} \cdot 2 \cos t dt = 4 \int_0^{\frac{\pi}{2}} \frac{\cos^3 t}{\sin t + \cos t} dt$$

由公式 $\int_0^{\frac{\pi}{2}} f(\sin x) dx = \int_0^{\frac{\pi}{2}} f(\cos x) dx$ 得

$$\int_0^{\frac{\pi}{2}} \frac{\cos^3 t}{\sin t + \cos t} dt = \int_0^{\frac{\pi}{2}} \frac{\sin^3 t}{\cos t + \sin t} dt$$

所以

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \frac{4-x^2}{x+\sqrt{4-x^2}} dx &= 4 \int_0^{\frac{\pi}{2}} \frac{\cos^3 t}{\sin t + \cos t} dt = 4 \cdot \frac{1}{2} \left[\int_0^{\frac{\pi}{2}} \frac{\cos^3 t}{\sin t + \cos t} dt + \int_0^{\frac{\pi}{2}} \frac{\sin^3 t}{\cos t + \sin t} dt \right] \\ &= 2 \int_0^{\frac{\pi}{2}} \frac{\sin^3 t + \cos^3 t}{\sin t + \cos t} dt = 2 \int_0^{\frac{\pi}{2}} (\sin^2 t - \sin t \cos t + \cos^2 t) dt \\ &= 2 \int_0^{\frac{\pi}{2}} \sin^2 t dt - 2 \int_0^{\frac{\pi}{2}} \sin t \cos t dt + 2 \int_0^{\frac{\pi}{2}} \cos^2 t dt = 2 \cdot \frac{1}{2} \cdot \frac{\pi}{2} - 1 + 2 \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \pi - 1 \end{aligned}$$

$$(3) \quad \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos^2 x}{x(\pi-2x)} dx;$$

解 利用公式 $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$ 得

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos^2 x}{x(\pi-2x)} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos^2 \left(\frac{\pi}{6} + \frac{\pi}{3} - x \right)}{\left(\frac{\pi}{6} + \frac{\pi}{3} - x \right) \left(\pi - 2 \left(\frac{\pi}{6} + \frac{\pi}{3} - x \right) \right)} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^2 x}{x(\pi-2x)} dx$$

所以

$$\begin{aligned} \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos^2 x}{x(\pi-2x)} dx &= \frac{1}{2} \left[\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos^2 x}{x(\pi-2x)} dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^2 x}{x(\pi-2x)} dx \right] = \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{x(\pi-2x)} dx \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \left(\frac{1}{x} + \frac{2}{\pi-2x} \right) dx = \frac{1}{2} \left[\frac{1}{\pi} \ln x - \frac{1}{\pi} \ln(\pi-2x) \right] \Big|_{\frac{\pi}{6}}^{\frac{\pi}{3}} = \frac{1}{\pi} \ln 2 \end{aligned}$$

$$(4) \quad \int_0^{\pi} \frac{1}{1+a \cos x} dx \quad (0 < a < 1);$$

解 利用公式 $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ 得

$$\int_0^{\pi} \frac{1}{1+a \cos x} dx = \int_0^{\pi} \frac{1}{1+a \cos(\pi-x)} dx = \int_0^{\pi} \frac{1}{1-a \cos x} dx$$

所以

$$\begin{aligned}\int_0^\pi \frac{1}{1+a\cos x} dx &= \frac{1}{2} \left[\int_0^\pi \frac{1}{1+a\cos x} dx + \int_0^\pi \frac{1}{1-a\cos x} dx \right] \\ &= \int_0^\pi \frac{1}{1-a^2\cos^2 x} dx = \int_0^\pi \frac{1}{\sin^2 x + (1-a^2)\cos^2 x} dx \\ &= \int_0^\pi \frac{1}{1+(1-a^2)\cot^2 x} \cdot \frac{1}{\sin^2 x} dx = -\int_0^\pi \frac{1}{1+(1-a^2)\cot^2 x} d\cot x\end{aligned}$$

(注意 $\frac{1}{\sin^2 x}$ 和 $\cot x$ 在 $x=0, \pi$ 处无界!!)

$$\begin{aligned}&= -\frac{1}{\sqrt{1-a^2}} \int_0^\pi \frac{1}{1+(\sqrt{1-a^2}\cot x)^2} d(\sqrt{1-a^2}\cot x) \\ &= -\frac{1}{\sqrt{1-a^2}} \arctan(\sqrt{1-a^2}\cot x) \Big|_0^\pi = -\frac{1}{\sqrt{1-a^2}} \left(-\frac{\pi}{2} - \frac{\pi}{2} \right) = \frac{\pi}{\sqrt{1-a^2}}\end{aligned}$$

(但需注意, 下面的方法是错误的)

$$\begin{aligned}\int_0^\pi \frac{1}{1+a\cos x} dx &= \int_0^\pi \frac{1}{1-a^2\cos^2 x} dx = \int_0^\pi \frac{1}{\sin^2 x + (1-a^2)\cos^2 x} dx \\ &= \int_0^\pi \frac{1}{\tan^2 x + 1-a^2} \cdot \frac{1}{\cos^2 x} dx = \int_0^\pi \frac{1}{\tan^2 x + (\sqrt{1-a^2})^2} d\tan x \\ &= \frac{1}{\sqrt{1-a^2}} \arctan \frac{\tan x}{\sqrt{1-a^2}} \Big|_0^\pi = 0\end{aligned}$$

结果是错的, 原因是在积分的过程中出现了 $\frac{1}{\cos^2 x}$ 和 $\tan x$, 它们在 $x = \frac{\pi}{2}$ 处是无界的,

如果想使用这样方法, 也可以, 但必须在 $x = \frac{\pi}{2}$ 处加分点将积分拆开, 这样代入积分限 $\frac{\pi}{2}$ 时便考虑了敛散性)

$$(5) \int_0^\pi \ln(1+\tan x) dx.$$

解 利用公式 $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ 得

$$\begin{aligned}\int_0^\pi \ln(1+\tan x) dx &= \int_0^\pi \ln \left(1 + \tan \left(\frac{\pi}{4} - x \right) \right) dx = \int_0^{\frac{\pi}{4}} \ln \left(1 + \frac{1-\tan x}{1+\tan x} \right) dx \\ &= \int_0^{\frac{\pi}{4}} \ln \frac{2}{1+\tan x} dx = \int_0^{\frac{\pi}{4}} \ln 2 dx - \int_0^{\frac{\pi}{4}} \ln(1+\tan x) dx = \frac{\pi}{4} \ln 2 - \int_0^{\frac{\pi}{4}} \ln(1+\tan x) dx\end{aligned}$$

解得

$$\int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx = \frac{\pi}{8} \ln 2$$

题 8 需利用公式

设 $f(x+T) = f(x)$ (周期函数), 则

$$(1) \int_a^{a+T} f(x) dx = \int_0^T f(x) dx = \int_{\frac{T}{2}}^{\frac{T}{2}+T} f(x) dx;$$

$$(2) \int_a^{a+nT} f(x) dx = n \int_0^T f(x) dx \quad (n \text{ 为正整数}).$$

8. 计算 $\int_{e^{-2n\pi}}^1 \left[\cos\left(\ln \frac{1}{x}\right) \right]' dx.$

解 注意到

$$\cos\left(\ln \frac{1}{x}\right) = \cos(-\ln x) = \cos \ln x$$

所以

$$\begin{aligned} \int_{e^{-2n\pi}}^1 \left[\cos\left(\ln \frac{1}{x}\right) \right]' dx &= \int_{e^{-2n\pi}}^1 |(\cos \ln x)'| dx = \int_{e^{-2n\pi}}^1 |(-\sin \ln x) \cdot \frac{1}{x}| dx \\ &= \int_{e^{-2n\pi}}^1 |\sin \ln x| \cdot \frac{1}{x} dx = \int_{e^{-2n\pi}}^1 |\sin \ln x| \cdot (\ln x)' dx = \int_{e^{-2n\pi}}^1 |\sin \ln x| d \ln x \end{aligned}$$

令 $t = \ln x$, 则 $d \ln x = dt$, 当 $x = e^{-2n\pi}$ 时 $t = -2n\pi$ (积分下限), 当 $x = 1$ 时 $t = 0$ (积分上限), 所以

$$\begin{aligned} \int_{e^{-2n\pi}}^1 \left[\cos\left(\ln \frac{1}{x}\right) \right]' dx &= \int_{e^{-2n\pi}}^1 |\sin \ln x| d \ln x = \int_{-2n\pi}^0 |\sin t| dt \quad (\text{被积函数周期为 } \pi) \\ &= 2n \int_0^{\pi} |\sin t| dt \quad (\text{利用公式 } \int_a^{a+nT} f(x) dx = n \int_0^T f(x) dx) \\ &= 2n \int_0^{\pi} \sin t dt = 2n(-\cos t) \Big|_0^{\pi} = 4n \end{aligned}$$

9. 设 $f(x) = \int_1^x \frac{\ln x}{1+x} dx$, 求 $f(x) + f\left(\frac{1}{x}\right).$

解 $f(x) + f\left(\frac{1}{x}\right) = \int_1^x \frac{\ln x}{1+x} dx + \int_1^{\frac{1}{x}} \frac{\ln x}{1+x} dx$

因为

$$\int_1^x \frac{\ln x}{1+x} dx = \int_1^x \frac{\ln t}{1+t} dt \stackrel{t=\frac{1}{u}}{=} \int_1^x \frac{\ln \frac{1}{u}}{1+\frac{1}{u}} \left(-\frac{1}{u^2} du \right) = \int_1^x \frac{\ln u}{u(1+u)} du = \int_1^x \frac{\ln x}{x(1+x)} dx$$

所以

$$\begin{aligned} f(x) + f\left(\frac{1}{x}\right) &= \int_1^x \frac{\ln x}{1+x} dx + \int_1^x \frac{\ln x}{x(1+x)} dx = \int_1^x \frac{\ln x}{x} dx = \int_1^x \frac{\ln t}{t} dt \\ &= \int_1^x \ln t d \ln t = \frac{1}{2} \ln^2 t \Big|_{t=1}^{t=x} = \frac{1}{2} \ln^2 x \end{aligned}$$

10. 设 $f(x) = \begin{cases} x e^{-x^2}, & x \geq 0 \\ \frac{1}{1+\cos x}, & -1 \leq x < 0 \end{cases}$, 计算 $\int_1^4 f(x-2) dx$.

解 令 $u = x - 2$, 则 $x = u + 2, dx = du$, 当 $x = 1$ 时 $u = -1$ (积分下限), 当 $x = 4$ 时 $u = 2$ (积分上限), 所以

$$\begin{aligned} \int_1^4 f(x-2) dx &= \int_{-1}^2 f(u) du = \int_{-1}^2 f(x) dx \\ &= \int_{-1}^0 \frac{1}{1+\cos x} dx + \int_0^2 x e^{-x^2} dx = \int_{-1}^0 \frac{1}{\cos^2 \frac{x}{2}} d\left(\frac{x}{2}\right) - \frac{1}{2} \int_0^2 e^{-x^2} d(-x^2) \\ &= \tan \frac{x}{2} \Big|_{-1}^0 - \frac{1}{2} e^{-x^2} \Big|_0^2 = \tan \frac{1}{2} - \frac{1}{2} e^{-4} + \frac{1}{2} \end{aligned}$$

11. 设 $f(x) = \begin{cases} 2x + \frac{3}{2}x^2, & -1 \leq x < 0 \\ \frac{x e^x}{(1+e^x)^2}, & 0 \leq x \leq 1 \end{cases}$, 求 $F(x) = \int_{-1}^x f(t) dt$ 的表达式.

解 当 $-1 \leq x < 0$ 时, 有

$$F(x) = \int_{-1}^x f(t) dt = \int_{-1}^x \left(2t + \frac{3}{2}t^2 \right) dt = \left(t^2 + \frac{1}{2}t^3 \right) \Big|_{-1}^x = \frac{1}{2}x^3 + x^2 - \frac{1}{2}$$

当 $0 \leq x \leq 1$ 时, 有

$$\begin{aligned}
F(x) &= \int_{-1}^x f(t)dt = \int_{-1}^0 f(t)dt + \int_0^x \frac{te^t}{(1+e^t)^2} dt \\
&= F(0) + \int_0^x \frac{t}{(1+e^t)^2} (1+e^t)' dt = -\frac{1}{2} + \int_0^x t \cdot \frac{1}{(1+e^t)^2} d(1+e^t) \\
&= -\frac{1}{2} + \int_0^x t d\left(-\frac{1}{1+e^t}\right) = -\frac{1}{2} + t\left(-\frac{1}{1+e^t}\right)\Big|_0^x - \int_0^x \left(-\frac{1}{1+e^t}\right) dt \\
&= -\frac{1}{2} - \frac{x}{1+e^x} + \int_0^x \left(1 - \frac{e^t}{1+e^t}\right) dt = -\frac{1}{2} - \frac{x}{1+e^x} + [t - \ln(1+e^t)]_0^x \\
&= -\frac{1}{2} - \frac{x}{1+e^x} + x - \ln(1+e^x) + \ln 2 = \frac{xe^x}{1+e^x} - \ln(1+e^x) - \frac{1}{2} + \ln 2
\end{aligned}$$

因此

$$F(x) = \begin{cases} \frac{1}{2}x^3 + x^2 - \frac{1}{2}, & -1 \leq x < 0 \\ \frac{xe^x}{1+e^x} - \ln(1+e^x) - \frac{1}{2} + \ln 2, & 0 \leq x \leq 1 \end{cases}$$

顺便指出, 因为 $f(x)$ 在区间 $[-1, 1]$ 上连续, 所以 $F(x)$ 是 $f(x)$ 在区间 $[-1, 1]$ 上的一个原函数. 如果将 $f(x)$ 换成

$$f(x) = \begin{cases} 2x + \frac{3}{2}x^2 + 1, & -1 \leq x < 0 \\ \frac{xe^x}{(1+e^x)^2}, & 0 \leq x \leq 1 \end{cases}$$

则 $f(x)$ 在 $x=0$ 处有跳跃间断点, 这时算出的 $F(x)$ 在区间 $[-1, 1]$ 上连续, 但需注意 $F(x)$ 就不是 $f(x)$ 在区间 $[-1, 1]$ 上的原函数了 (因为 $F(x)$ 在 $x=0$ 处不可导).

12. 设 $f(x) = x, x \geq 0, g(x) = \begin{cases} \sin x, & 0 \leq x \leq \frac{\pi}{2} \\ 0, & x > \frac{\pi}{2} \end{cases}$, 分别求 $0 \leq x \leq \frac{\pi}{2}$ 与 $x > \frac{\pi}{2}$ 时积

分 $\int_0^x f(t)g(x-t)dt$ 的表达式.

解 令 $u = x - t$, 则 $t = x - u, dt = -du$, 当 $t = 0$ 时 $u = x$ (积分下限), 当 $t = x$ 时 $u = 0$ (积分上限), 于是

$\int_0^x f(t)g(x-t)dt = \int_x^0 f(x-u)g(u)(-du) = \int_0^x f(x-u)g(u)du$
 (注意, $g(x)$ 是分段函数, 这里通过定积分换元法对 $g(x-t)$ 进行简化)

当 $0 \leq x \leq \frac{\pi}{2}$ 时, 由于 $0 \leq u \leq x$, 即 $x-u \geq 0$, 所以

$$\begin{aligned}\int_0^x f(t)g(x-t)dt &= \int_0^x f(x-u)g(u)du = \int_0^x (x-u)\sin u du \\ &= x \int_0^x \sin u du - \int_0^x u \sin u du = x(-\cos u)\Big|_0^x + (u \cos u - \sin u)\Big|_0^x = x - \sin x\end{aligned}$$

当 $x > \frac{\pi}{2}$ 时, 由于 $0 \leq u \leq x$, 即 $x-u \geq 0$, 所以

$$\begin{aligned}\int_0^x f(t)g(x-t)dt &= \int_0^x f(x-u)g(u)du = \int_0^{\frac{\pi}{2}} (x-u)\sin u du + \int_{\frac{\pi}{2}}^x (x-u) \cdot 0 du \\ &= x \int_0^{\frac{\pi}{2}} \sin u du - \int_0^{\frac{\pi}{2}} u \sin u du + 0 = x(-\cos u)\Big|_0^{\frac{\pi}{2}} + (u \cos u - \sin u)\Big|_0^{\frac{\pi}{2}} = x - 1\end{aligned}$$

13. 计算积分 $\int_0^{n\pi} x|\sin x|dx$ ($n=1,2,\dots$).

解 记 $a_k = \int_{(k-1)\pi}^{k\pi} x|\sin x|dx$ ($k=1,2,\dots,n$), 当 k 是奇数时, 有

$$\begin{aligned}a_k &= \int_{(k-1)\pi}^{k\pi} x|\sin x|dx = \int_{(k-1)\pi}^{k\pi} x \sin x dx = (-x \cos x + \sin x)\Big|_{(k-1)\pi}^{k\pi} \\ &= -k\pi \cos k\pi - [-(k-1)\pi \cos(k-1)\pi] = -k\pi(-1)^k + (k-1)\pi(-1)^{k-1} \\ &= k\pi + (k-1)\pi = (2k-1)\pi\end{aligned}$$

当 k 是偶数时, 有

$$\begin{aligned}a_k &= \int_{(k-1)\pi}^{k\pi} x|\sin x|dx = -\int_{(k-1)\pi}^{k\pi} x \sin x dx = -(-x \cos x + \sin x)\Big|_{(k-1)\pi}^{k\pi} \\ &= k\pi \cos k\pi + [-(k-1)\pi \cos(k-1)\pi] = k\pi(-1)^k - (k-1)\pi(-1)^{k-1} \\ &= k\pi + (k-1)\pi = (2k-1)\pi\end{aligned}$$

于是无论 k 是偶数还是奇数均有 $a_k = (2k-1)\pi$, 因此

$$\begin{aligned}\int_0^{n\pi} x|\sin x|dx &= \int_0^{\pi} x|\sin x|dx + \int_{\pi}^{2\pi} x|\sin x|dx + \dots + \int_{(n-1)\pi}^{n\pi} x|\sin x|dx \\ &= a_1 + a_2 + \dots + a_n = \pi + 3\pi + \dots + (2n-1)\pi = \pi n^2\end{aligned}$$

14. 已知 $f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 2-x, & 1 < x \leq 2 \end{cases}$, 计算 $S_n = \int_{2n}^{2n+2} f(x-2n)e^{-x}dx$ ($n=0,1,\dots$),

并计算极限 $\lim_{n \rightarrow \infty} (S_0 + S_1 + \cdots + S_n)$.

解

$$\begin{aligned}
 S_0 &= \int_0^2 f(x) \mathbf{e}^{-x} dx = \int_0^1 x \mathbf{e}^{-x} dx + \int_1^2 (2-x) \mathbf{e}^{-x} dx \\
 &= -\left. (x+1) \mathbf{e}^{-x} \right|_0^1 - \left. (1-x) \mathbf{e}^{-x} \right|_1^2 = 1 - \frac{2}{\mathbf{e}} + \frac{1}{\mathbf{e}^2} = \left(1 - \frac{1}{\mathbf{e}}\right)^2 \\
 S_n &= \int_{2n}^{2n+2} f(x-2n) \mathbf{e}^{-x} dx \stackrel{u=x-2n}{=} \int_0^2 f(u) \mathbf{e}^{-u-2n} du = \int_0^2 f(u) \mathbf{e}^{-u} \mathbf{e}^{-2n} du \\
 &= \mathbf{e}^{-2n} \int_0^2 f(u) \mathbf{e}^{-u} du = \mathbf{e}^{-2n} S_0 = \left(1 - \frac{1}{\mathbf{e}}\right)^2 \mathbf{e}^{-2n}
 \end{aligned}$$

利用所得结果, 有

$$\begin{aligned}
 \lim_{n \rightarrow \infty} (S_0 + S_1 + \cdots + S_n) &= \lim_{n \rightarrow \infty} (S_0 + \mathbf{e}^{-2} S_0 + \cdots + \mathbf{e}^{-2n} S_0) \\
 &= \lim_{n \rightarrow \infty} \frac{1 - (\mathbf{e}^{-2})^{n+1}}{1 - \mathbf{e}^{-2}} S_0 = \frac{1}{1 - \mathbf{e}^{-2}} S_0 = \frac{1}{1 - \mathbf{e}^{-2}} \left(1 - \frac{1}{\mathbf{e}}\right)^2 = \frac{\mathbf{e} - 1}{\mathbf{e} + 1}
 \end{aligned}$$